

SBC 201 Chapter 25 Gypsum Board, Gypsum Panel Products and Plaster — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–12 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 25, source file Reference\SBC 201 2024\source_reference\Chapter_25 – GYPSUM BOARD, GYPSUM PANEL PRODUCTS AND PLASTER.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **72** independently checkable numeric records (**53** Verified, **19** Verify source). Flattened Tables 2508.6 and 2512.6 and clause **2508.6.4** (175 mm mm) are listed in the register and are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, gypsum-assembly listing, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapter 7, Chapter 16, Section 110.3.6, Sections 1210.2 / 1210.3, Sections 1403.2 / 1404.4, Section 2211.1.1, Section 2306.3, SBC 301 Sections 12.2.1 / 13.5.6, ASTM C842 / C840 / C926 / C1063 / C1280, GA-216, commentary examples, bibliography edition years, or the existing chapter summary. Where Chapter 25 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Automatic sprinkler protection is unconfirmed. Chapter 25 does not branch gypsum or plaster millimetres on NFPA 13 versus 13R.
4. Construction type is unconfirmed. A typical R-2 tower is Type I or II concrete or steel; wood-frame gypsum shear walls, wood-joint gypsum diaphragm ceilings and reinforced gypsum concrete roof decks are not assumed and are omitted from Sections 4–12.
5. Exterior cement plaster (stucco), interior three-coat plaster, exposed-aggregate finish, wood furring and studless partitions are unconfirmed product choices. Those rows are Conditional.
6. Riyadh is treated as a dry (B) climate-zone basis for 2510.6.1. Moist (A) / marine (C) 2510.6.2 millimetres and drainage-efficiency percentages are omitted unless a coastal or moist-zone site is opened in the gap register.
7. Tables 2506.2, 2507.2, 2508.1, 2509.2 and 2511.1.1 are material/standard name maps. They are cited by standard designation; no millimetre is invented from those tables.
8. Fire-resistance-rated gypsum fastening, joint treatment beyond 2508.5, and plaster coat thickness in ASTM C842 / C926 cannot be concluded from this chapter extract alone.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, finish system, backing, climate-zone branch or exception exists.
Not typical	Unrelated occupancy-only or wood-frame/roof-deck rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 25 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 25 source text.
Verify source	OCR, flattened table, page-split, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 25 application	Required project action
Wall and ceiling finish system	Unconfirmed: gypsum board vs interior plaster vs cement plaster vs exposed aggregate	Selects 2508 board rules, 2511 interior plaster coats, 2512 exterior stucco, or 2513 bedding-coat geometry	Issue a wall-type and ceiling-type schedule by unit, corridor, stair, lobby and podium elevation
Exterior cladding	Unconfirmed whether cement plaster/stucco is used on the shaft or podium	Turns on 2510 lath/WRB, 2512 weep screed, coat counts and curing	Freeze stucco versus other Chapter 14 coverings before detailing lath, WRB and weep screed
Wet rooms vs high-humidity amenities	Dwelling showers and water closets expected; sauna, steam room, gang shower and indoor pool not assumed	2509.2 / Table 2509.2 backer and 2509.3 bans; 2511.5 moisture barrier on wood-frame splash walls. Commentary sauna/pool examples are not adopted as code	Mark every tub, shower, WC and any pool/spa room on the finish schedule; do not use water-resistant gypsum backing over a vapor retarder in shower/tub compartments
Wood furring / wood framing	Unconfirmed; tower studs typically cold-formed steel	2504.1.1 50 mm least dimension and the 25 mm by 50 mm solid-backing exception apply only to wood supports	Use the wood-support row only where wood stripping or furring is specified
Studless plaster partitions	Unconfirmed; not assumed for the tower	2504.1.2 50 mm minimum thickness	Do not use studless plaster partitions without a non-load-bearing check against Chapter 16 deflection
Wood-frame gypsum shear	Not assumed on a Type I/II tower	2505.1 sends design to 2306.3 and seismic limits to SBC 301 12.2.1 . No millimetres in this chapter	If a wood podium shear wall is added, extract 2306.3; do not size gypsum shear from this matrix
Wood-joint gypsum diaphragm ceiling	Not assumed	2508.6 and Table 2508.6 (flattened) plus 2508.6.4 OCR 175 mm mm. Omitted from Sections 4–12	If a wood-joint ceiling diaphragm is added, re-open 2508.6 against a clean table; do not adopt concatenated shear values

Decision / gap	Current project basis	Why it controls Chapter 25 application	Required project action
Reinforced gypsum concrete roof	Not assumed	2514.2 50 mm / 40 mm / 800 mm / 2000 Pa omitted from Sections 4–12	If a gypsum-concrete deck is proposed, re-filter 2514; do not copy commentary ASTM C317 Class A/B strengths
Climate zone for stucco WRB	Riyadh treated as dry (B); not locked from SBC 601/602	2510.6.1 dry two-layer 10-minute or 60-minute paths apply over wood-based sheathing. 2510.6.2 5 mm / 90 percent moist-marine extras are omitted	Confirm the energy-code climate zone if stucco is used over wood sheathing; do not apply A/C drainage millimetres on a dry-B basis
Sprinkler / mixed use / storeys	Unconfirmed	Chapter 25 has no occupancy, sprinkler or storey branch for these installation millimetres	Do not wait on NFPA 13 vs 13R to apply 2508 wet-area and 2512 weep-screed checks
Rated gypsum assemblies	Unconfirmed listings	2502.1 / 2506.2 / 2507.2 send fire protection to Chapter 7 ; 2508.5 joint treatment follows the tested assembly	Take fastening, layers and listings from the Chapter 7 tested design, not from this chapter’s commentary
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	Rated shafts, corridors and stairs cannot be concluded from Chapter 25 alone	Engage the fire/code consultant before freezing gypsum type (regular, Type X, glass-mat) on rated walls

4. Scope, fire-rated gypsum and inspection

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2501.1	Chapter gypsum and plaster scope	This chapter governs materials, design, construction and quality of gypsum board, gypsum panel products, lath, gypsum plaster, cement plaster and reinforced gypsum concrete	Those materials used on the building	2501.2 other approved wall or ceiling coverings	Direct	Treat Chapter 25 as the gypsum/plaster installation chapter, not the fire-rating or interior-finish-class chapter	Verified
2501.2	Other wall or ceiling coverings	Other approved wall or ceiling coverings shall be installed in accordance with the manufacturer’s recommendations and the conditions of approval	Non-gypsum/non-plaster wall or ceiling coverings proposed	This chapter does not regulate those other materials	Conditional	Send metal, masonry, wood or vinyl coverings to their own chapters; do not force them through 2508–2512	Verified
2502.1	Fire-protection overlay	Lathing, plastering and gypsum board and gypsum panel product construction shall use this chapter’s manner and materials and, when required for fire protection, shall also comply with Chapter 7	Gypsum or plaster used for fire protection	Chapter 7 ratings, listings and fastening are not in this extract	External verification	Match the tested Chapter 7 assembly for every shaft, stair, corridor and unit-separation membrane; do not invent fire-resistance hours here	Verified
2503.1	Lath and gypsum inspection	Lath, gypsum board and gypsum panel products shall be inspected in accordance with Section 110.3.6	Lath, gypsum board or gypsum panel product installation	Inspection timing in 110.3.6 is not imported	External verification	Hold concealment of lath and gypsum until the 110.3.6 inspection point is shown on the permit inspection list	Verified
2506.1	Gypsum identification and storage	Gypsum board, gypsum panel products and accessories shall be identified by the manufacturer’s designation to indicate compliance with the standards in this section and stored to protect such materials from the weather	Gypsum board, panel products and accessories on site	None stated	Direct	Require mill labels on delivered board and a dry storage note on the specification; do not accept unmarked bundles	Verified
2506.2	Gypsum material standards	Gypsum board and gypsum panel products shall conform to Table 2506.2 and Chapter 35 and, where required for fire protection, to Chapter 7	Gypsum board and panel products	Table 2506.2 is a standard-name map; no millimetre is taken from the flattened table	External verification	Put ASTM C1396 / C1178 / C1177 (as applicable) on the product specification; take fire-rated type from the Chapter 7 listing	Verified
2507.1–2507.2	Lath and plaster standards	Lathing and plastering materials shall be manufacturer-marked, weather-protected, and shall conform to Table 2507.2 and Chapter 35 and, where required for fire protection, to Chapter 7	Lath, plaster and related accessories	Table 2507.2 is a standard-name map	External verification	Name ASTM C847 / C933 / C1032 / C150 (as applicable) on the plaster specification; do not import mix proportions from the table	Verified
2508.1	Gypsum construction installation standards	Gypsum board, gypsum panel products and gypsum plaster shall be of the materials in Tables 2506.2 and 2507.2 , assembled and installed to Tables 2508.1 and 2511.1.1 and Chapter 35	Gypsum board, panel and gypsum plaster construction	Installation millimetres in GA-216, ASTM C840, C1280, C842 and C926 are not imported	External verification	Cite GA-216 or ASTM C840 for board hanging and ASTM C1280 for exterior gypsum sheathing; do not copy fastener spacing from commentary	Verified

5. Support of gypsum and plaster

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2504.1.1	Wood support least dimension	Wood supports for lath, gypsum board or gypsum panel products, and wood stripping or furring, shall be not less than 50 mm nominal thickness in the least dimension	Wood supports, stripping or furring for gypsum or lath	Exception: wood furring strips over solid backing shall be not less than 25 mm by 50 mm nominal	Conditional	If wood furring is used on concrete or masonry, specify 25 × 50 mm over solid backing; otherwise keep wood supports at 50 mm least dimension	Verified
2504.1.2	Studless plaster partition thickness	Vertically erected studless solid plaster partitions of 10 mm and 20 mm rib metal lath, 12.5 mm -thick gypsum lath, gypsum board or gypsum panel product shall be not less than 50 mm thick	Studless solid plaster partitions	None stated in this section. Lateral design remains in Chapter 16	Conditional	Do not detail a studless plaster partition thinner than 50 mm ; keep it non-load-bearing and check Chapter 16 deflection separately	Verified

6. Weather, wet rooms and tile backer

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2508.2	Exterior and wet-use gypsum limits	Gypsum wallboard or gypsum plaster shall not be used on any exterior surface where that gypsum construction is exposed directly to the weather. Gypsum wallboard shall not be used where there will be direct exposure to water or continuous high humidity. Gypsum sheathing on exterior surfaces shall be installed in accordance with ASTM C1280	Exterior gypsum or gypsum in wet/high-humidity exposure	Weather-exposed-surface definition is in Section 202, not quantified here. C1280 thicknesses are not imported	Direct	Keep paper-faced gypsum wallboard and gypsum plaster off weather-exposed faces; specify ASTM C1280 gypsum sheathing only as a concealed exterior substrate	Verified
2508.2.1	Weather protection before install	Gypsum wallboard, gypsum lath or gypsum plaster shall not be installed until weather protection for the installation is provided	Gypsum wallboard, lath or plaster being installed	None stated. Commentary “one week” heating is not adopted	Direct	Sequence gypsum hang after the enclosure is weathertight; do not hang interior board on an open elevation	Verified
2509.1	Shower and public-toilet walls	Showers and public toilet walls shall conform to Section 1210.2	Showers and public toilet walls	1210.2 finish materials and heights are not imported	External verification	Send unit showers and any public toilet rooms to 1210.2 for the nonabsorbent finish; use 2509.2 only for the backer	Verified
2509.2	Tub and shower tile backer	Materials used as a base for wall tile in tub and shower areas and wall and ceiling panels in shower areas shall be of materials listed in Table 2509.2 , installed per the manufacturer’s recommendations	Tub and shower wall tile; shower wall and ceiling panels	Table 2509.2 names glass-mat gypsum backing ASTM C1178 ; nonasbestos fiber-cement ASTM C1288 or ISO 8336, Category C ; nonasbestos fiber-mat cementitious backer ASTM C1325 . No millimetre is taken from the table	Direct	Specify a Table 2509.2 backer on every tub/shower wall and shower ceiling; do not use regular paper-faced gypsum as the shower tile base	Verified
2509.2	Water-closet tile backer	Water-resistant gypsum backing board shall be used as a base for tile in water closet compartment walls, installed in accordance with GA-216 or ASTM C840 and the manufacturer’s recommendations. Regular gypsum wallboard is permitted under tile or wall panels in other wall and ceiling areas when installed to GA-216 or ASTM C840	Tile on water-closet compartment walls; tile or panels in other rooms	Regular gypsum wallboard is not the 2509.2 shower/tub backer	Direct	Use water-resistant gypsum backing behind WC-compartment tile; keep regular board for dry rooms only	Verified
2509.3	Water-resistant backing bans	Water-resistant gypsum backing board shall not be used over a vapor retarder in shower or bathtub compartments, and shall not be used where there will be direct exposure to water or in areas subject to continuous high humidity	Water-resistant gypsum backing board	None stated	Direct	Omit a vapor retarder behind green board in shower/tub walls; use a Table 2509.2 cement or glass-mat backer wherever the finish is in the water stream	Verified
2511.5	Wood-frame splash-wall barrier	Showers and public toilet walls shall conform to Sections 1210.2 and 1210.3 . Where wood-frame walls and partitions are covered on the interior with cement plaster or tile of similar material and are subject to water splash, the framing shall be protected with an approved moisture barrier	Wood-frame walls with interior cement plaster or tile, subject to water splash	1210.2 / 1210.3 values are not imported	Conditional	If any wet room is wood-framed, show an approved moisture barrier on the framing before tile or cement plaster	Verified

7. Gypsum board installation

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2508.3	Board edges and contact	Edges and ends of gypsum board and gypsum panel products shall occur on framing members, except edges and ends perpendicular to framing. Edges and ends shall be in moderate contact except in concealed spaces where fire-resistance-rated construction, shear resistance or diaphragm action is not required	Gypsum board and gypsum panel products	Concealed non-rated, non-shear, non-diaphragm spaces may omit moderate contact. Rated, shear and diaphragm edges still bear on framing	Direct	Land parallel edges on studs or joists; keep joints in moderate contact on shafts, stairs, unit separations and any shear wall	Verified
2508.3.1	Floating-angle fasteners	Fasteners at top and bottom plates of vertical assemblies, or at edges and ends of horizontal assemblies perpendicular to supports, and at the wall line, may be omitted except on shear-resisting elements or fire-resistance-rated assemblies. Fasteners shall not fracture the face paper with the fastener head	Gypsum board on walls or ceilings	Floating angles are not permitted on shear elements or fire-resistance-rated assemblies	Direct	Allow floating angles only on non-rated, non-shear partitions and ceilings; screw every edge on rated shafts, stairs and corridor walls	Verified
2508.4	Ceiling adhesive fastening	Gypsum board and gypsum panel products secured to framing with adhesives in ceiling assemblies shall be attached using an approved fastening schedule. Expandable foam adhesives shall conform to ASTM D6464 . Other adhesives shall conform to ASTM C557	Adhesive-attached gypsum on ceilings	None stated	Conditional	If ceiling board is adhesive-hung, require a fastening schedule plus ASTM D6464 or C557 on the specification; do not rely on adhesive alone	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2508.5	Rated-assembly joint treatment	Gypsum board and gypsum panel product fire-resistance-rated assemblies shall have joints and fasteners treated	Fire-resistance-rated gypsum assemblies	Exception: treatment not required where (1) a decorative finish equivalent to joint treatment is applied; (2) single-layer joints occur over wood framing; (3) square-edge, tongue-and-groove, backing board or sheathing; (4) multilayer joints of adjacent layers are offset; or (5) the assembly was tested without joint treatment	Direct	Treat joints and fasteners on every rated assembly unless the listed UL/GA design is one of the five numbered exceptions	Verified

8. Cement-plaster lath, backing and dry-climate WRB

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2510.3	Cement-plaster installation standards	Installation of exterior and interior cement plaster and lathing materials shall comply with ASTM C926 and ASTM C1063	Cement plaster (stucco) and its lath/furring	Coat thicknesses in those ASTM standards are not imported	External verification	Name ASTM C926 and C1063 on every stucco specification; do not copy scratch/brown/finish millimetres from commentary	Verified
2510.4	Corrosion-resistant lath	Metal lath and lath attachments shall be of corrosion-resistant material	Metal lath and attachments in cement plaster	None stated	Conditional	Specify galvanized or equivalent corrosion-resistant lath, screws, screeds and control joints on the stucco elevation	Verified
2510.5–2510.5.1	Lath rigidity and edge support	Backing or lath shall provide sufficient rigidity to permit plaster application. Where lath on vertical surfaces extends between rafters or similar projecting members, solid backing shall be installed to support the lath and attachments	Cement plaster lath, including at eaves/rafter projections	2510.5.3: wire backing is not required under expanded metal lath or paper-backed wire fabric lath	Conditional	Show solid backing at lath edges where the wall plane is interrupted by rafters or similar members; omit extra wire backing under expanded metal or paper-backed fabric lath	Verified
2510.5.2.1	Gypsum as cement-plaster backing	Gypsum lath or gypsum wallboard shall not be used as a backing for cement plaster	Cement plaster over gypsum lath or wallboard	Exception: gypsum lath or wallboard is permitted, with a water-resistive barrier, as backing for self-furred metal lath or self-furred wire fabric lath and cement plaster on (1) horizontal supports of ceilings or roof soffits, or (2) interior walls	Conditional	Do not plaster cement stucco directly to gypsum wallboard on exterior walls; if gypsum is used on an interior wall or soffit, add a WRB and self-furred lath	Verified
2510.5.2.2	Gypsum sheathing under stucco	Gypsum sheathing is permitted as a backing for metal or wire fabric lath and cement plaster on walls. A water-resistive barrier shall be provided in accordance with Section 2510.6	Cement plaster over gypsum sheathing on walls	WRB details follow 2510.6 / 1403.2	Conditional	If gypsum sheathing backs exterior stucco, show the 2510.6 WRB on the wall section; do not omit the barrier	Verified
2510.6	Stucco water-resistive barrier	Water-resistive barriers shall be installed as required in Section 1403.2 and, where applied over wood-based sheathing, shall comply with 2510.6.1 or 2510.6.2	Exterior plaster WRB; additional rules over wood-based sheathing	1403.2 and 1404.4 flashing millimetres are not imported. 2510.6.2 moist/marine extras are omitted on the Riyadh dry-B basis	External verification	Show a Chapter 14 WRB behind stucco; if the sheathing is wood-based, use the 2510.6.1 dry-climate options below	Verified
2510.6.1	Dry-climate stucco WRB over wood	For dry (B) climate zones, either: (1) two layers of 10-minute Grade D paper, or water resistance equal to or greater than two layers of WRB complying with ASTM E2556 Type I, each layer a separate continuous plane, with 1404.4 flashing directed between the layers; or (2) 60-minute Grade D paper, or water resistance equal to or greater than one layer of ASTM E2556 Type II, separated from the stucco by foam plastic insulating sheathing, another nonwater-absorbing layer, or a drainage space	Stucco WRB over wood-based sheathing in a dry (B) climate zone	Item 1 and Item 2 are alternative paths. Do not import 1404.4 flashing sizes	Conditional	On a dry-B stucco wall over wood sheathing, pick either the two-layer 10-minute / E2556 I path or the 60-minute / E2556 II path with a non-absorbing separator or drainage space	Verified
2510.7	Masonry and concrete plaster bond	Surfaces shall be clean, free from efflorescence, sufficiently damp and rough for proper bond. If insufficiently rough, approved bonding agents or a Portland cement dash-bond coat mixed in proportions of not more than two parts volume of sand to one part volume of Portland cement or plastic cement shall be applied. The dash-bond coat shall be left undisturbed and moist-cured not less than 24 hours	Cement plaster applied to masonry or concrete	None stated	Conditional	If stucco or interior cement plaster goes on concrete or masonry, specify a bonding agent or 2:1 dash coat with 24-hour moist cure before the base coat	Verified

9. Interior plaster

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2511.1	Interior plaster coat count	Gypsum plaster or cement plaster shall be not less than three coats over metal lath or wire fabric lath and not less than two coats over other bases permitted by this chapter	Interior gypsum plaster or cement plaster	Exception: gypsum veneer plaster and cement plaster specifically designed and approved for one-coat applications. Commentary veneer-thickness caps are not adopted	Conditional	Specify three-coat plaster on metal or wire lath and two-coat on solid bases unless a listed one-coat veneer product is used	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2511.1.1	Interior plaster installation table	Installation of lathing and plaster materials shall conform to Table 2511.1.1 and Section 2507	Interior plaster	Table 2511.1.1 names ASTM C842, C843, C841, C1063, C926, C754 and C1007. Thicknesses in those standards are not imported	External verification	Cite the matching ASTM installation standard on the interior-plaster specification; do not invent coat millimetres from this matrix	Verified
2511.2	Interior plaster base bans	Plaster shall not be applied directly to fiber insulation board. Cement plaster shall not be applied directly to gypsum lath or gypsum plaster except as specified in 2510.5.1 and 2510.5.2	Interior plaster over fiberboard or gypsum plaster base	Cement plaster over gypsum only via the 2510.5.1 / 2510.5.2 backing rules	Conditional	Keep plaster off fiber insulation board; do not scratch-coat cement plaster onto gypsum lath without the 2510.5 WRB and self-furred lath exception	Verified
2511.3	Plaster grounds thickness	Where installed, grounds shall ensure the minimum thickness of plaster as set forth in ASTM C842 and ASTM C926 . Plaster thickness shall be measured from the face of lath and other bases	Interior plaster grounds	ASTM thickness tables are not imported	External verification	Set grounds from ASTM C842 or C926 measured from the face of the lath; do not dimension plaster thickness from commentary figures	Verified
2511.4	Concrete-ceiling plaster thickness	Surface condition as 2510.7. Approved specially prepared gypsum plaster for concrete, or approved acoustical plaster, is permitted. Total thickness of base-coat plaster on concrete ceilings shall be as set forth in ASTM C842 or ASTM C926 . If more than that maximum is required, metal lath or wire fabric lath shall be installed before plastering	Interior plaster on concrete ceilings	ASTM maximum thicknesses are not imported	Conditional	If a concrete soffit is plastered thicker than the ASTM ceiling cap, add metal or wire lath before the extra plaster	Verified

10. Exterior cement plaster

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2512.1	Exterior cement-plaster coat count	Cement plaster shall be not less than three coats over metal lath, wire fabric lath or gypsum-board backing as 2510.5, and not less than two coats over masonry or concrete. If the plaster surface is completely covered by veneer or facing, or completely concealed by another wall, two coats are permitted provided the total thickness is as set forth in ASTM C926	Exterior cement plaster	Concealed/veneered two-coat path still takes ASTM C926 total thickness (not imported)	Conditional	Specify three-coat stucco on lath or gypsum backing and two-coat on masonry/concrete; use the concealed two-coat option only with an ASTM C926 thickness submittal	Verified
2512.1.1	Slab-on-grade plaster termination	On wood-frame or steel-stud construction with an on-grade concrete floor slab, exterior plaster shall cover, but not extend below, the lath and paper. Application of lath, paper and flashing or drip greens shall comply with ASTM C1063	Exterior plaster on stud walls at a slab-on-grade	ASTM C1063 drip/flashing profiles are not imported. The extract token “drip greens” is not repaired	Conditional	Stop stucco at the lath/paper line above the slab; detail the weep screed and flashing to ASTM C1063 rather than burying paper below the plaster	Verified
2512.1.2	Exterior weep screed	A minimum 0.5 mm (No. 26 galvanized sheet gage), corrosion-resistant weep screed with a minimum vertical attachment flange of 90 mm shall be provided at or below the foundation plate line on exterior stud walls in accordance with ASTM C926. Place the weep screed not less than 100 mm above the earth or 50 mm above paved areas, of a type that drains trapped water to the exterior. The WRB shall lap the attachment flange. Exterior lath shall cover and terminate on that flange	Exterior cement plaster on stud walls	None stated	Conditional	Draw a No. 26 / 0.5 mm weep screed with 90 mm flange, 100 mm above grade or 50 mm above paving; lap WRB over the flange and terminate lath on it	Verified
2512.2	Plaster plasticity agents	Only approved plasticity agents and approved amounts shall be added to Portland or blended cements. Additional lime or plasticizers shall not be added where plastic cement or masonry cement is used. Hydrated lime or equivalent lime putty as a plasticizer shall not exceed ASTM C926	Cement plaster mix	ASTM C926 lime limits are not imported	External verification	Ban extra lime in plastic or masonry cement mixes; cap any added lime at the ASTM C926 amount on the mix design	Verified
2512.3	Exterior gypsum plaster ban	Gypsum plaster shall not be used on exterior surfaces	Any exterior plaster finish	None stated	Direct	Specify cement plaster (stucco) for exterior plaster; keep gypsum plaster inside	Verified
2512.4	Cement-plaster cold-weather limits	Plaster coats shall be protected from freezing for not less than 24 hours after set. Plaster shall be applied when the ambient temperature is higher than 4°C , unless provisions are made to keep cement plaster work above 4°C during application and 48 hours thereafter	Exterior cement plaster application	Heated enclosure permitted in lieu of ambient 4°C	Conditional	Add a cold-weather plaster note: no freeze for 24 hours after set; hold the work above 4°C during placement and for 48 hours	Verified
2512.5	Second-coat flatness	The second coat shall be brought out to proper thickness, riddled and floated sufficiently rough to bond the finish coat. The second coat shall not have variations greater than 6.5 mm in any direction under a 1500 mm straightedge	Exterior cement plaster brown/second coat	None stated	Conditional	Require a 1500 mm straightedge check with 6.5 mm maximum deviation on the second coat before the finish coat	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Table 2512.6	Cement-plaster curing table	First and second coats shall be applied and moist-cured as set forth in ASTM C926 and Table 2512.6	Exterior cement plaster curing and coat intervals	Table 2512.6 is flattened HTML (concatenated headers). No 48-hour / 7-day cell is adopted. 2512.7 allows the second coat over gypsum backing or unit masonry as soon as the first coat is hard enough. 2512.8 alternate method may omit first-coat curing	Conditional	Hold stucco curing times until Table 2512.6 is read from a clean code page; do not schedule 48-hour or 7-day intervals from this extract	Verify source
2512.8.1	Alternate-method aluminate cap	Where using the 2512.8 alternate method, calcium aluminate cement up to 15 percent of the weight of the Portland cement is permitted to be added to the mix	Alternate two-coat application under 2512.8	Applies only on that method. Second coat still cured to ASTM C926 and Table 2512.6	Conditional	If the contractor uses the alternate scratch-to-brown sequence, cap calcium aluminate at 15 percent of the Portland cement weight	Verified
2512.9	Finish-coat bond and cover	Cement plaster finish coats shall be applied over base coats that have been in place for the time periods in ASTM C926 . The third or finish coat shall be applied with sufficient material and pressure to bond and cover the brown coat and shall be thick enough to conceal the brown coat	Exterior cement plaster finish coat	ASTM C926 wait times are not imported	External verification	Apply the finish coat only after the ASTM C926 base-coat interval; the brown coat must be fully concealed	Verified

11. Exposed aggregate plaster

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2513.2	Exposed-aggregate hardness	Aggregate shall be marble chips, pebbles or similar durable, moderately hard (three or more on the Mohs hardness scale), nonreactive materials, applied manually or mechanically	Exposed-aggregate plaster finish	None stated	Conditional	Specify aggregate Mohs 3 or harder and nonreactive on the finish schedule if an exposed-chip plaster is used	Verified
2513.3	Bedding-coat proportions and strength	Interior or exterior cement bedding coat: one part Portland cement and one part Type S lime; or one part blended cement and one part Type S lime; or masonry cement; or plastic cement; and a maximum of three parts graded white or natural sand by volume. Interior gypsum bedding coat: 45 kg neat gypsum plaster and a maximum of 90 kg graded white sand. Factory-prepared bedding coats are permitted. Exterior bedding coat minimum compressive strength 6895 kPa	Exposed-aggregate bedding coat	Factory-prepared bedding coats may substitute for the site mix	Conditional	Put the 1:1:3 cement-lime-sand (max) mix or the 45 kg / 90 kg gypsum mix on the plaster spec; require 6895 kPa for exterior bedding	Verified
2513.4	Exposed-aggregate overall thickness	Bedding coat may be applied over the first (scratch) coat provided the ultimate overall thickness is a minimum of 22 mm , including lath. Over concrete or masonry, overall thickness shall be a minimum of 12.5 mm	Exposed-aggregate plaster	None stated	Conditional	Detail 22 mm total including lath on framed plaster, or 12.5 mm total on concrete or masonry	Verified
2513.7	Bedding-coat moist cure	Cement plaster base coats shall be cured in accordance with ASTM C926 . Cement plaster bedding coats shall retain sufficient moisture for hydration for 24 hours minimum or, where necessary, be kept damp for 24 hours by light water spraying	Cement plaster exposed-aggregate bedding coat	ASTM C926 base-coat cure is not imported. Do not substitute commentary Table 2512.6 interval language for this 2513.7 bedding-coat rule	Conditional	Keep the bedding coat moist for 24 hours ; still cure the scratch/brown coats to ASTM C926	Verified

12. Shear walls and suspended ceilings

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
2505.2	Cold-formed steel gypsum shear	Cold-formed steel-frame shear walls sheathed with gypsum board or gypsum panel products and constructed in accordance with Section 2211.1.1 are permitted to resist wind and seismic loads. Walls resisting seismic loads shall be subject to the limitations in Section 12.2.1 of SBC 301	CFS gypsum-sheathed shear walls	2211.1.1 capacities and SBC 301 12.2.1 system limits are not imported. Wood-frame gypsum shear (2505.1 / 2306.3) is omitted as not typical	Conditional	If gypsum is used as CFS shear sheathing, send the wall to 2211.1.1 and SBC 301 12.2.1; do not assign a shear value from this chapter	Verified
2506.2.1	Suspended acoustical ceiling standards	Metal suspension systems for acoustical and lay-in panel ceilings shall comply with ASTM C635 and Section 13.5.6 of SBC 301 for installation in high seismic areas	Metal-suspended acoustical or lay-in panel ceilings	ASTM C635 duty class and SBC 301 13.5.6 seismic restraint millimetres are not imported	Direct	Name ASTM C635 and SBC 301 13.5.6 on corridor, lobby and amenity lay-in ceilings; do not invent hanger or perimeter-gap millimetres here	Verified

13. Project-use controls

1. Use **Verified** rows for initial scoping after the row trigger is confirmed (gypsum board, wet room, stucco, plaster or lay-in ceiling actually present).
2. Do not use the Table **2512.6** row as a design-release curing schedule. The attached table is flattened; hold times stay **Verify source** until a clean code page is read.
3. Do not import Chapter 7 fire-resistance hours, 1210 wet-finish heights, 1403.2 / 1404.4 WRB and flashing millimetres, 2211.1.1 / 2306.3 shear values, SBC 301 12.2.1 / 13.5.6 seismic limits, or ASTM C842 / C926 / C840 / C1063 / C1280 / GA-216 thicknesses from this matrix.

4. Do not adopt commentary figures (3 mm paper, flame-spread 25, 18-gage / 150 mm wire backing, 6 mm veneer-plaster cap, ASTM C317 Class A/B MPa and density).
5. Do not apply 2510.6.2 moist/marine **5 mm / 90 percent** drainage on the Riyadh dry-B basis, and do not reopen wood-joist diaphragm Table 2508.6 or 2514 gypsum-concrete decks without a gap-register change.
6. Record gypsum type, stucco use, wet-room backer and WRB decisions in the project Golden Thread; this matrix is not evidence of SCD NOC or stamped compliance.

14. Coverage summary

Internal inventory of the attached Chapter 25 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 72
- **Verified:** 53
- **Verify source:** 19

Counts by top-level section

Top-level section	Records
2501	0
2502	0
2503	0
2504	7
2505	0
2506	0
2507	0
2508	17
2509	0
2510	8
2511	3
2512	22
2513	10
2514	5

Appended tables: 2506.2, 2507.2, 2508.1, 2508.6, 2509.2, 2511.1.1, 2512.6. Material/standard maps (2506.2, 2507.2, 2508.1, 2509.2, 2511.1.1) contain no independently checkable SI geometry. Numeric cells and footnotes in Tables **2508.6** and **2512.6** are Verify source.

Coverage cross-check against SBC 201 Chapter 25 Gypsum Board, Gypsum Panel Products and Plaster (2024)_CS.md was topics-only: chapter scope; Chapter 7 overlay; wet-area backer and 2509.3 bans; 1210 companion; exterior cement plaster, weep screed and dry-climate WRB; gypsum not a cement-plaster backing except 2510.5.2.1. No CS.md value was copied into a matrix cell.

15. Unresolved-source register

Affected table / clause	Why unverified	Control note
Table 2508.6	Flattened HTML: concatenated headers (MATERIALTHICKNESS...MIMUM FASTENER SIZE), concatenated body cells, OCR head dimension (~6 mm vs 6 mm). Thickness, spacing, shear (N/m) and fastener cells plus notes a (25 percent), b (50 percent) and c (No. 6 screws) were inventoried and not adopted	Wood-joist gypsum diaphragm ceilings are omitted from Sections 4–12. Do not reconstruct 12.5 mm / 400–600 mm / 1300–1000 N/m values from this extract
2508.6.4	Fastener spacing published as 175 mm mm (duplicated unit token). Edge distance 10 mm in the same sentence is unambiguous and was inventoried as Verified; the on-center limit is not	Same omitted-use control as Table 2508.6. Do not “correct” the spacing to 175 mm from memory
Table 2512.6	Flattened HTML: concatenated headers (COATMINIMUM PERIOD...). First/second-coat 48-hour and 7-day cells and notes a–c (24 hours / 24-hour / 48-hour) were inventoried and not adopted	Project-use curing row is Verify source . Hold the stucco schedule until a clean Table 2512.6 is read; 2512.4 24 hours / 4°C / 48 hours in running text remain Verified